CRICKET PLAYER PERFORMANCE TRACKING

Group Project

Submitted to Immaculate College for Women,

Chinnakanganankuppam, Cuddalore-06

In partial fulfillment for the award of the degree of

BACHELOR OF COMPUTER APPLICATION

Submitted by

Lakshmi M (23131210500122018)

Sharmila S (23131210500122041)

Sridharani R (23131210500122043)

Suguna S(23131210500122044)

Archana S(23131210500122003)

DEPARTMENT OF COMPUTER SCIENCE

IMMACULATE COLLEGE FOR WOMEN, CUDDALORE-607 006

(Affiliated to Annamalai University, Chidambaram)

Cricket Player Performance Tracking Application : CRICKSTATS

User Story:

As a Salesforce Admin, I want to create a comprehensive player performance tracking application within Salesforce to capture and analyze detailed match-by-match information of cricket players. This application will include custom objects such as
Player and Match Performance with carefully defined fields and relationships to ensure data integrity and accessibility. By leveraging master-detail and lookup relationships, the application will enable robust tracking of individual and team performances, facilitate insightful reporting and analytics, and support the overall management of player statistics, thereby enhancing strategic decision-making and player development initiatives.

1. INTRODUCTION

1.1 Project Overview :

CRICKSTATS is a Salesforce-based application designed to track, manage, and analyze cricket players' performance on a match-by-match basis. The system uses custom objects, relationships, dashboards, and automation tools to provide structured performance insights.

It enables coaches, analysts, and administrators to record player statistics, generate reports, and support strategic decision-making.

1.2 Purpose :

The main purpose of this project is:

To digitize player performance records

To improve accuracy and accessibility of cricket statistics To provide real-time analytics

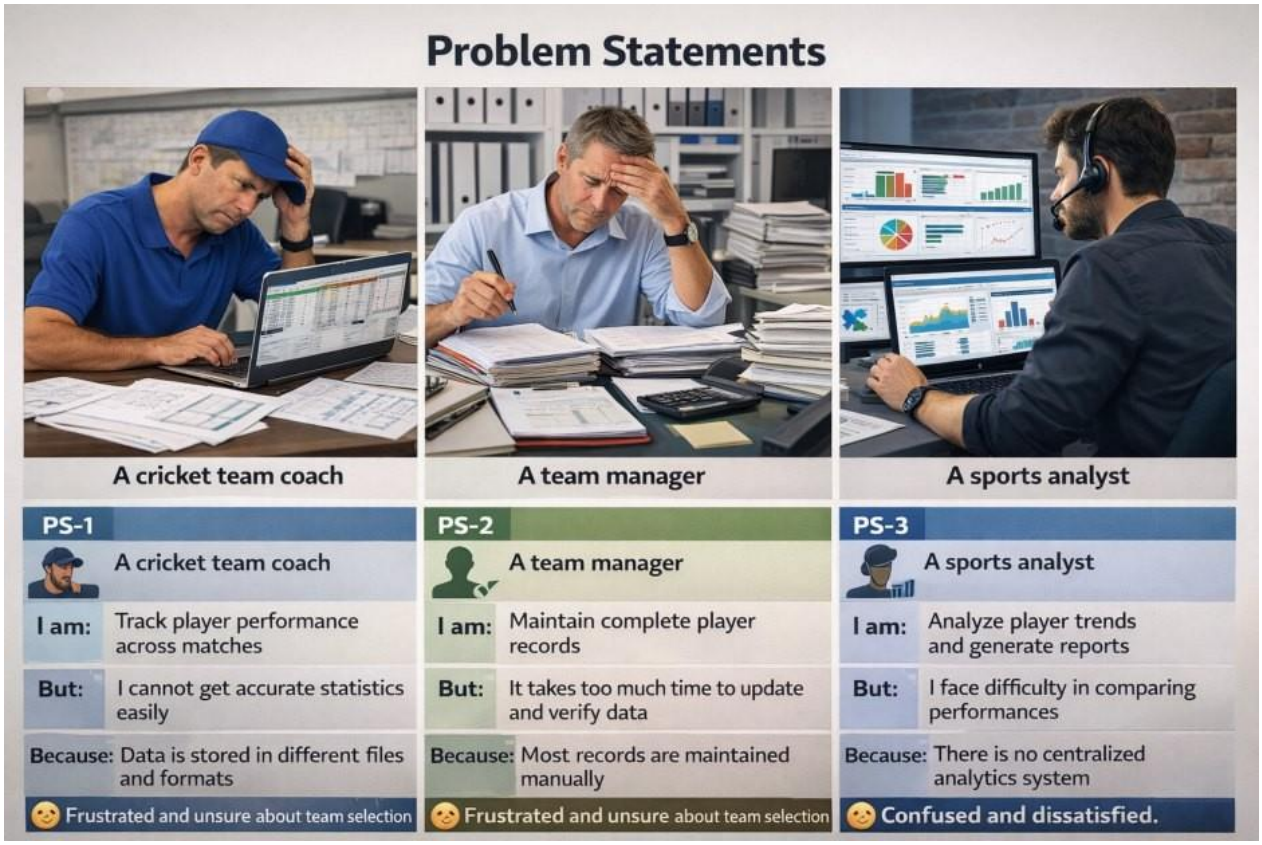
To support player development and team planning To automate data management using Salesforce

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A cricket team coach	Track player performance across matches	I cannot get accurate statistics easily	Data is stored in different files and formats	Frustrated and unsure about team selection

PS-2	A team manager	Maintain complete player records	It takes too much time to update and verify data	Most records are maintained manually	Stressed and inefficient
PS-3	A sports analyst	Analyze player trends and generate reports	I face difficulty in comparing performances	There is no centralized analytics system	Confused and dissatisfied



2.2 Empathy Map Canvas

Target Users

Team Managers
Coaches Analysts
Salesforce Admins

Empathy Analysis

Aspect Description

Think & Feel: Need accurate data, better insights

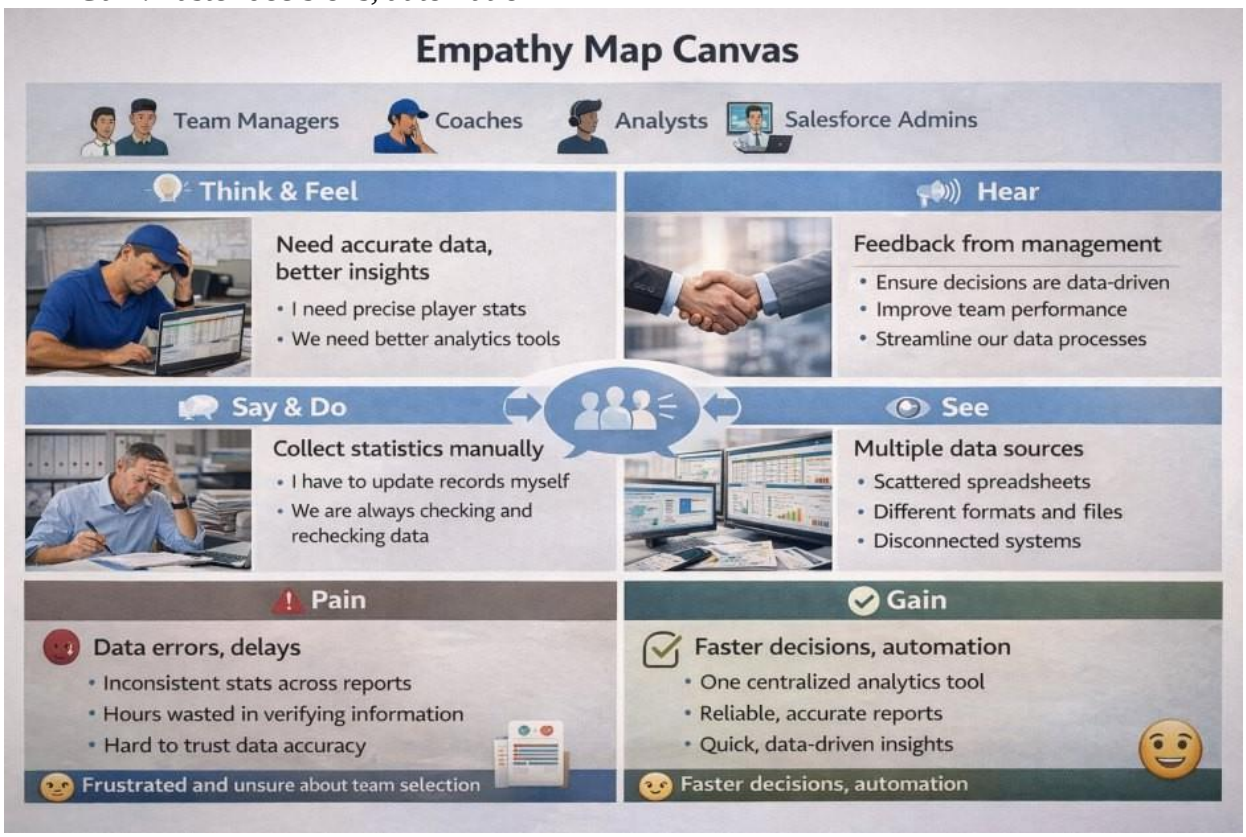
Hear: Feedback from management

See: Multiple data sources

Say & Do: Collect statistics manually

Pain: Data errors, delays

Gain: Faster decisions, automation



2.3 Brainstorming

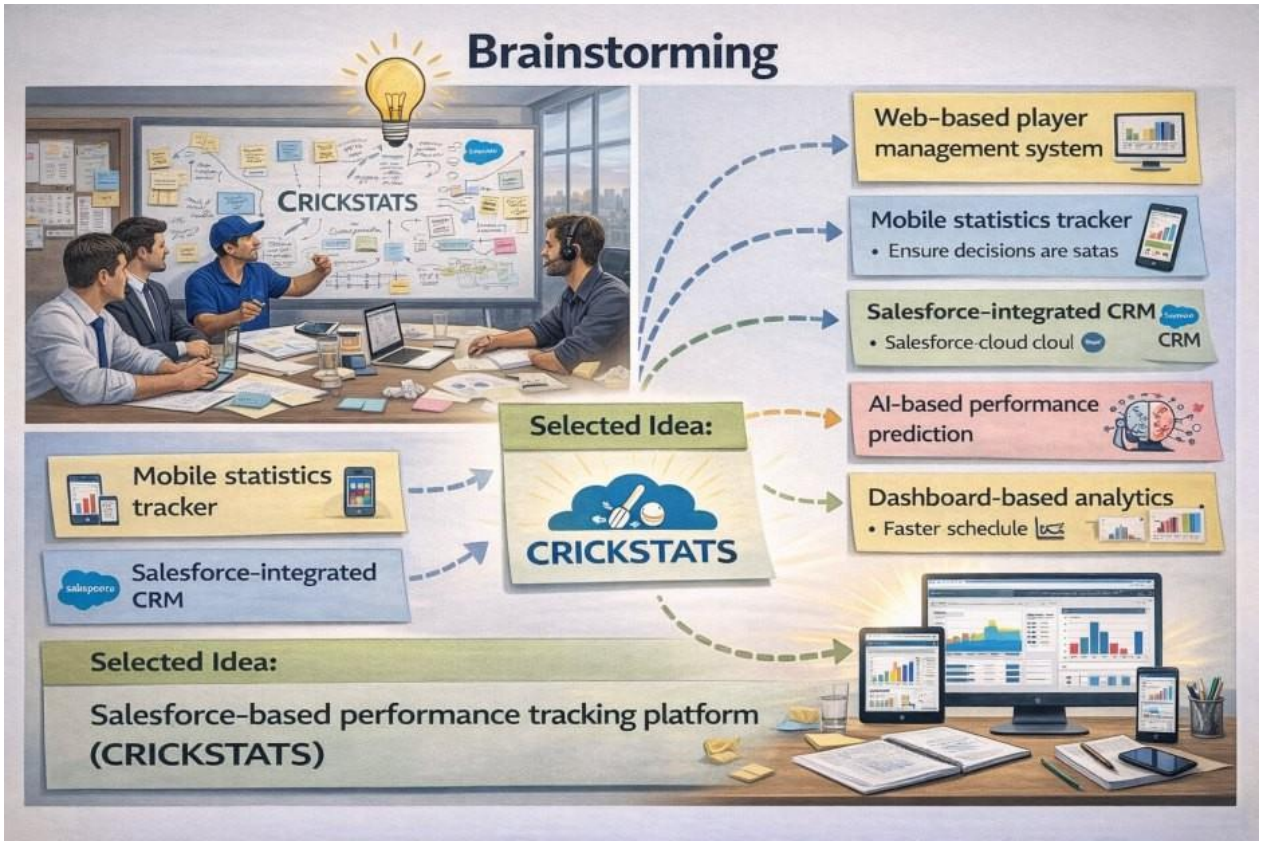
Brainstormed Ideas

Web-based player management system
Mobile statistics tracker
Salesforce-integrated CRM

AI-based performance prediction

Dashboard-based analytics

Selected Idea: Salesforce-based performance tracking platform (CRICKSTATS)



3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Journey Stages:

Login

Enter Player Data

Record Match Performance

Generate Reports

Analyze Results

Decision Making

3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Player Registration
FR-2	Match Performance Entry
FR-3	Dashboard Generation
FR-4	Report Creation
FR-5	Flow Automation

Non-Functional Requirements(NFRs)-CRICKSTATS NFR-1:Usability

The system should be easy to understand and use by coaches, managers, analysts, and admins without requiring technical training.

Requirements

Simple and intuitive user interface

Clear navigation menus

Easy-to-read dashboards

Minimal learning curve

Examples:

A new coach should be able to:
Log in
Enter match data
View player reports

within a few minutes without special training.

Example: The Salesforce Lightning interface provides drag-and-drop dashboards and

user-friendly forms.

NFR-2: Security

The system must protect sensitive player data from unauthorized access. Requirements

Role-based access control
Secure authentication

Encrypeted Data Storage

Regular backups

.

Examples:

Coaches can view player statistics

Managers can edit records Analysts

can generate reports

Only admins can modify system settings

Example: Salesforce Profiles and Permission Sets restrict access to confidential information.

NFR-3: Reliability

The system should function correctly without failure and store data accurately. Requirements

No data loss
Automatic backups

Error handling System

stability

Example:

If a manager enters match statistics and loses internet connection, the data should be

saved automatically when reconnected.

Salesforce cloud backup ensures recovery in case of system failure.

NFR-4: Performance

The system should respond quickly and handle large volumes of data efficiently.

Requirements

Fast page loading Quick

report generation

Smooth dashboard
rendering

Minimal response time

Example:

When an analyst generates a performance report for 100+ players, the report should load within 2–3 seconds.

Optimized Salesforce queries and caching improve response speed.

NFR-5: Availability

The system should be accessible whenever users need it.

Requirements

24/7 system availability
Minimal downtime
Cloud-based access
Disaster recovery

Example:

Coaches should be able to access CRICKSTATS even during tournaments at night or on weekends.

Salesforce provides high uptime with cloud hosting.

NFR-6: Scalability

The system should support increasing numbers of users, players, and records.

Requirements

Handle growing datasets Support
more teams and leagues Expand
storage capacity Maintain
performance under load

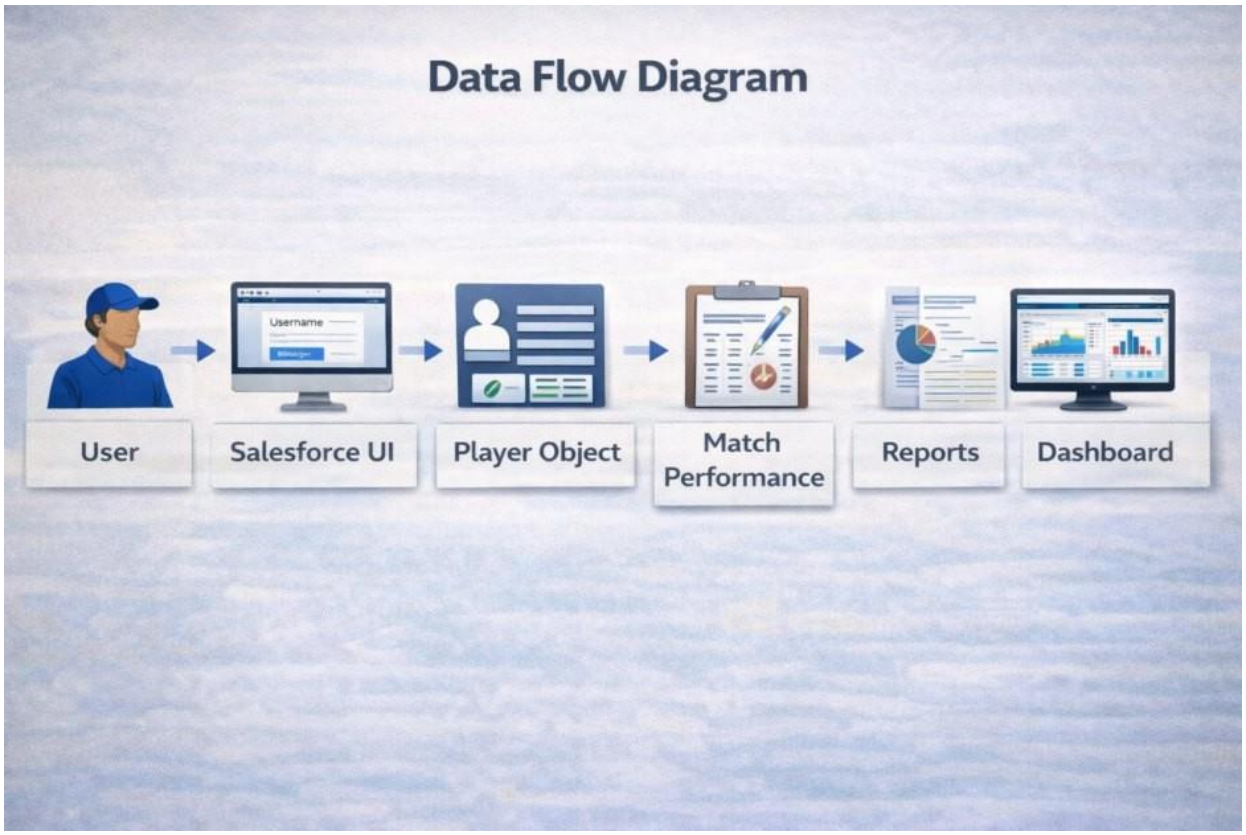
Example:
If CRICKSTATS is extended from one team to 50 teams, the system should still perform smoothly.

Salesforce cloud infrastructure automatically scales storage and processing.

3.3 Data Flow Diagram

Data Flow

User Salesforce UI Player Object Match Performance Reports Dashboard



3.4 Technology Stack

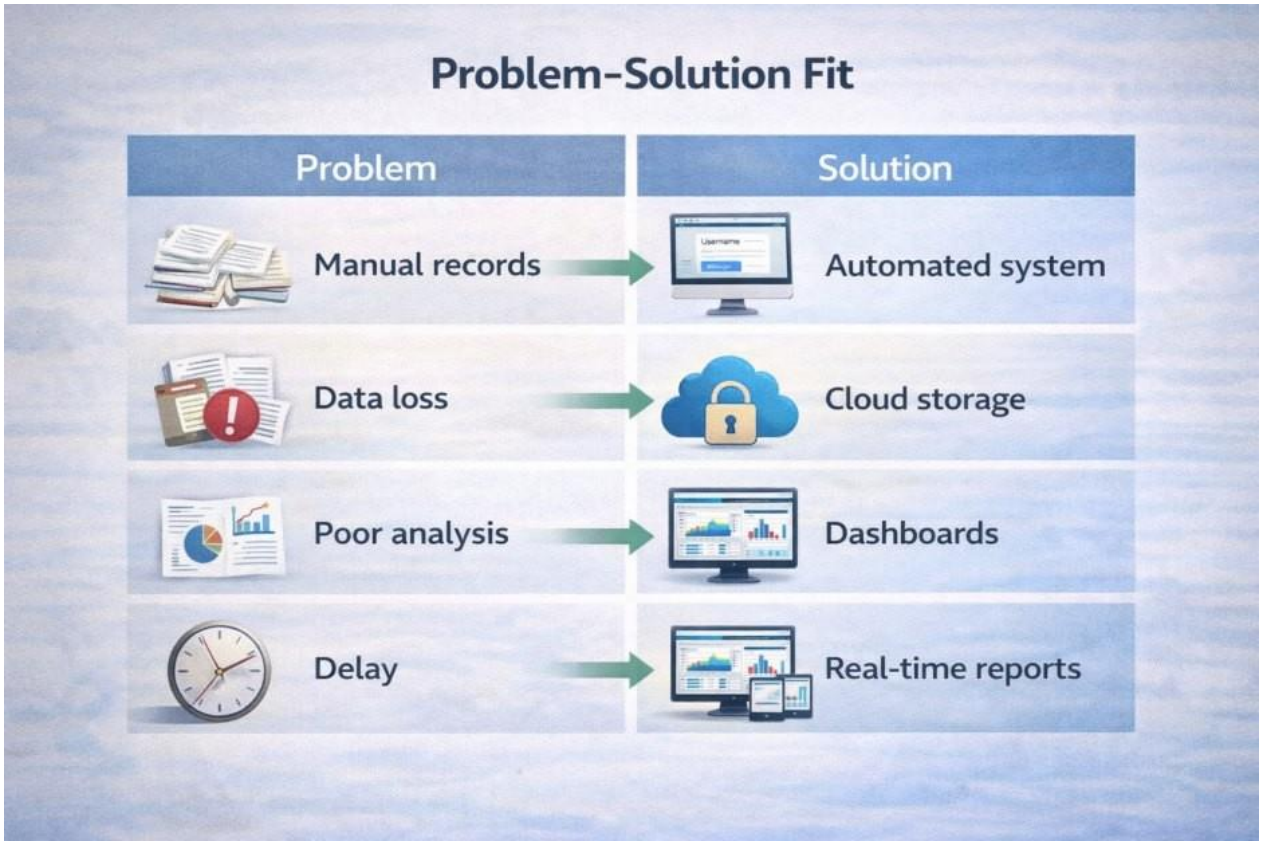
Layer	Technology
Frontend	Salesforce Lightning
Analytics	Reports & Dashboards
Automation	Salesforce Flows
Backend	Salesforce Objects
Database	Salesforce Cloud

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The CRICKSTATS system effectively addresses the key challenges faced in traditional cricket performance management. Earlier, most player and match data were maintained using manual records, which were time-consuming and prone to errors. This issue has been resolved by implementing an automated system through Salesforce, which ensures accurate and efficient data entry and management. The problem of data loss, caused by improper storage and lack of backups, is overcome by using secure cloud storage that safely stores all information and allows easy retrieval. Poor analysis of player performance, due to scattered and unorganized

data, is addressed through interactive dashboards that present clear visual insights and reports. Additionally, delays in accessing performance information and generating reports are eliminated by providing real-time reporting features, enabling users to make faster and more informed decisions. Thus, CRICKSTATS successfully bridges the gap between existing problems and modern technological solutions.



4.2 Proposed Solution

Description

CRICKSTATS provides:

- Custom objects (Player, Match Performance)
- Master-detail relationships
- Roll-up summaries
- Lightning App
- Automated flows
- Interactive dashboards

Key Features

- Centralized database

Live statistics
Secure access
Scalable design

Easy reporting

4.3 Solution Architecture

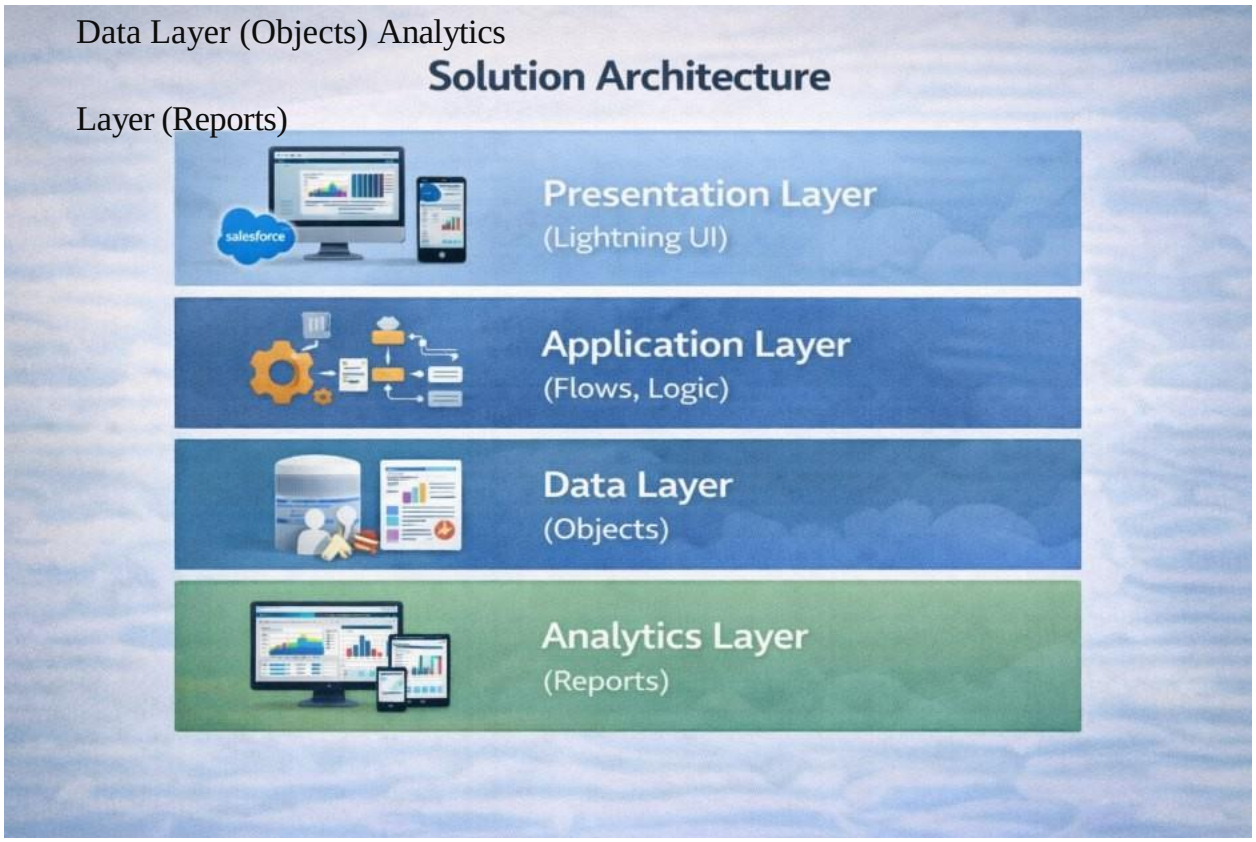
Architecture Layers

Presentation Layer (Lightning UI)

Application Layer (Flows, Logic)

Data Layer (Objects) Analytics

Layer (Reports)



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Agile-based execution

Object creation automation reporting

Sprint-wise development

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As an admin, I want to create a Salesforce developer account so that I can configure and deploy the CRICKSTATS system.	1	High	sridharani
Sprint-1	Data Modeling	USN-2	As an admin, I want to create Player and Match Performance objects so that player and match data can be stored systematically.	3	High	lakshmi
Sprint-2	Navigation	USN-3	As a user, I want separate tabs for each object so that I can easily navigate and manage records within the application.	3	High	suguna
Sprint-2	UI Design	USN-4	As a coach, I want a dedicated Lightning App so that all performance tracking features are accessible from one dashboard.	3	High	archana
Sprint-2	Data Structure	USN-5	As an admin, I want to create player and match fields so that complete performance details can be captured accurately.	5	Medium	sharmi
Sprint-2	Data Structure	USN-6	As a user, I want customized page layouts so that only relevant fields are displayed during data entry.	5	High	sridharani

Sprint-2	Automation	USN-7	As a system, I want to validate player and match inputs so that incorrect data entries are prevented.	3	High	lakshmi
Sprint-3	Automation	USN-8	As a system, I want automated flows to update player statistics so that performance data remains accurate.	5	High	suguna

Player of the match

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

6. Project Development Phase:

Project Flow

Milestone 1 : Creation of developer account

Milestone 2 : Object Creation

Milestone 3 : Tabs

Milestone 4 : The Lightning App

Milestone 5 : Fields & Relationships

Milestone 6 : Roll-up Summary

Milestone 7 : Flows

Implementation

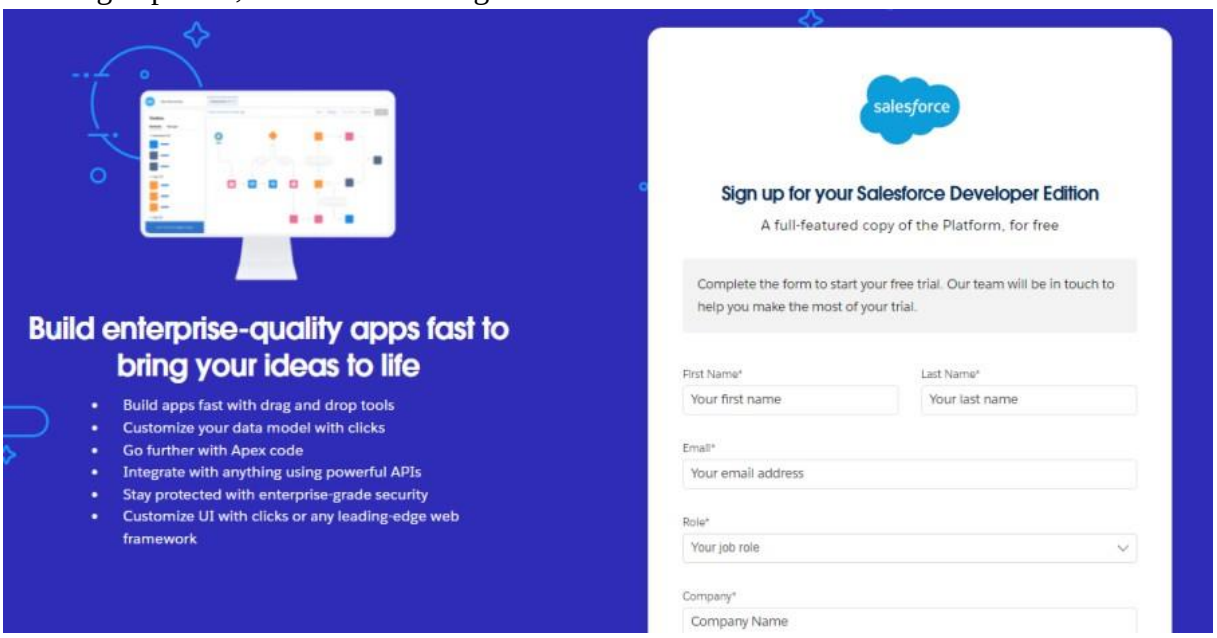
Milestone 1 - Salesforce developer account creation :

Activity 1: Creating Developer Account:

Creating a developer org in salesforce.

1. Go to https://developer.salesforce.com/signup

2. On the sign up form, enter the following details :



1) First name & Last name

2) Email

3) Role : Developer

4) Company : College or Company Name

5) County : India

6) Postal Code : pin code

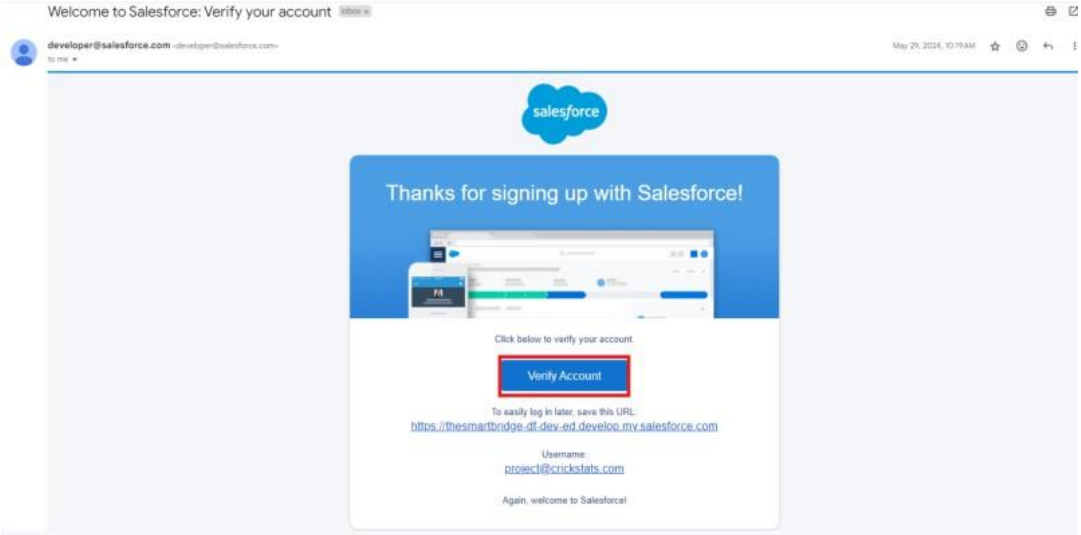
7) Username : should be a combination of your name and company This need not be an actual email id, you can give anything in that format :

username@organization.com

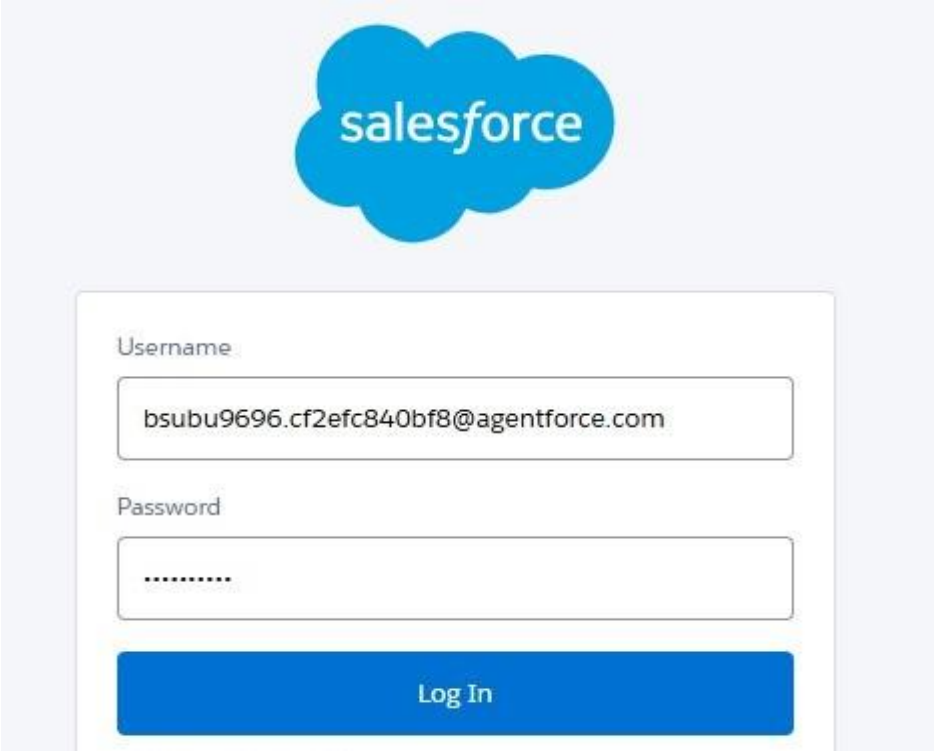
Click on sign me up after filling these.

Activity 2: Account Activation :

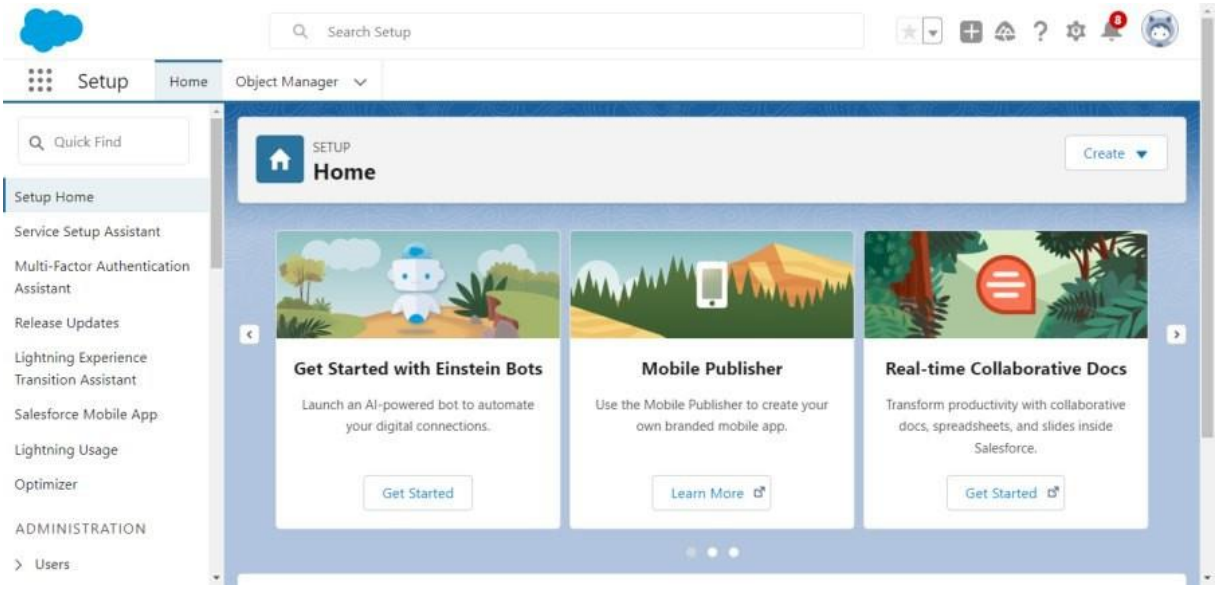
1. Go to the inbox of the email that you used while signing up. Click on the verify. account to activate your account. The email may take 5-10mins.
2. Click on Verify Account.



3. Give a password and answer a security question and click on change password.



4. Then you will redirect t o your salesforces

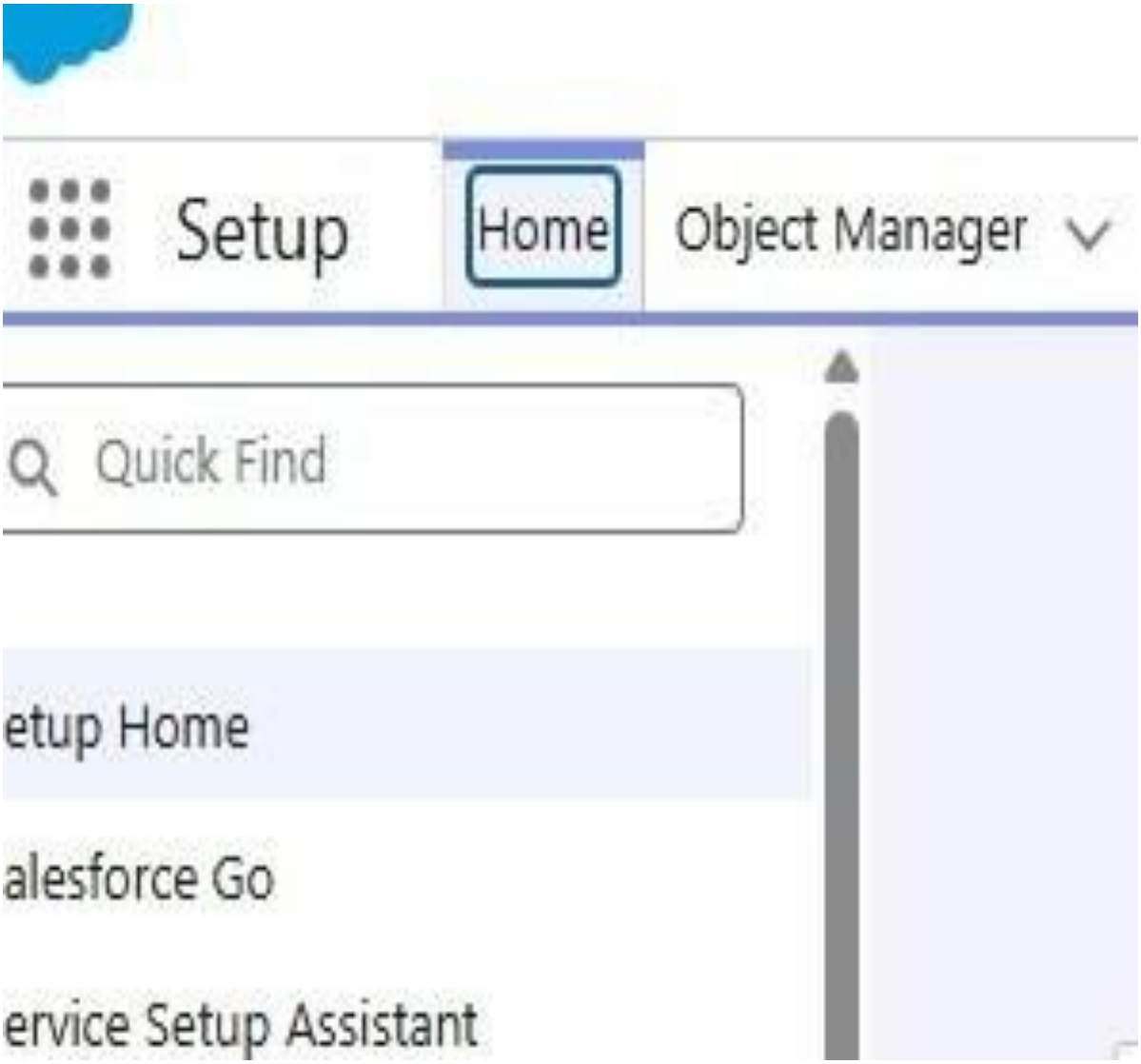


Milestone 2 - Object Creation Activity

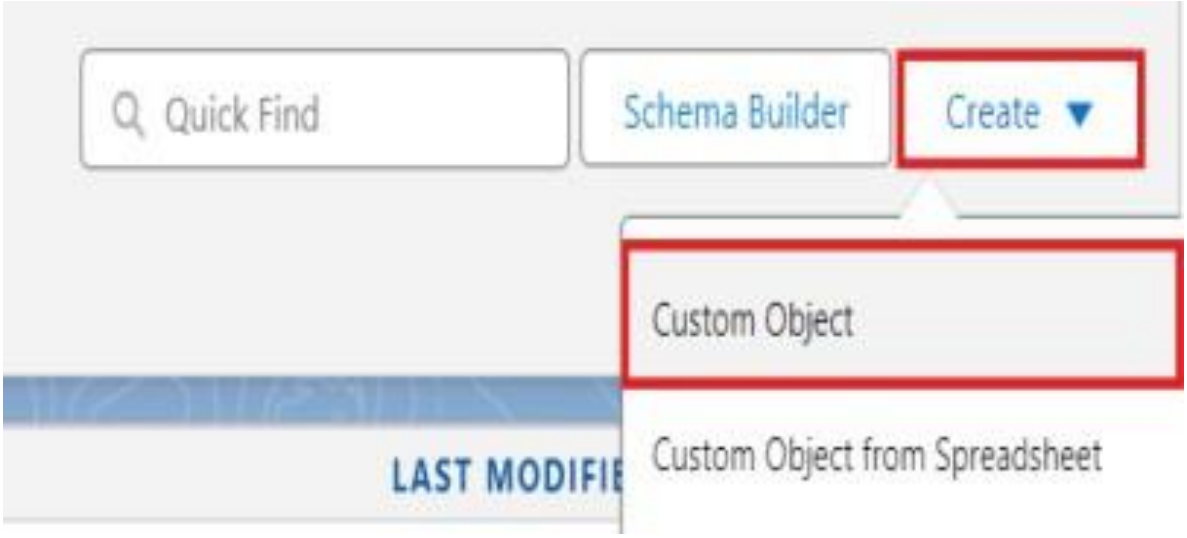
1 : Player Object

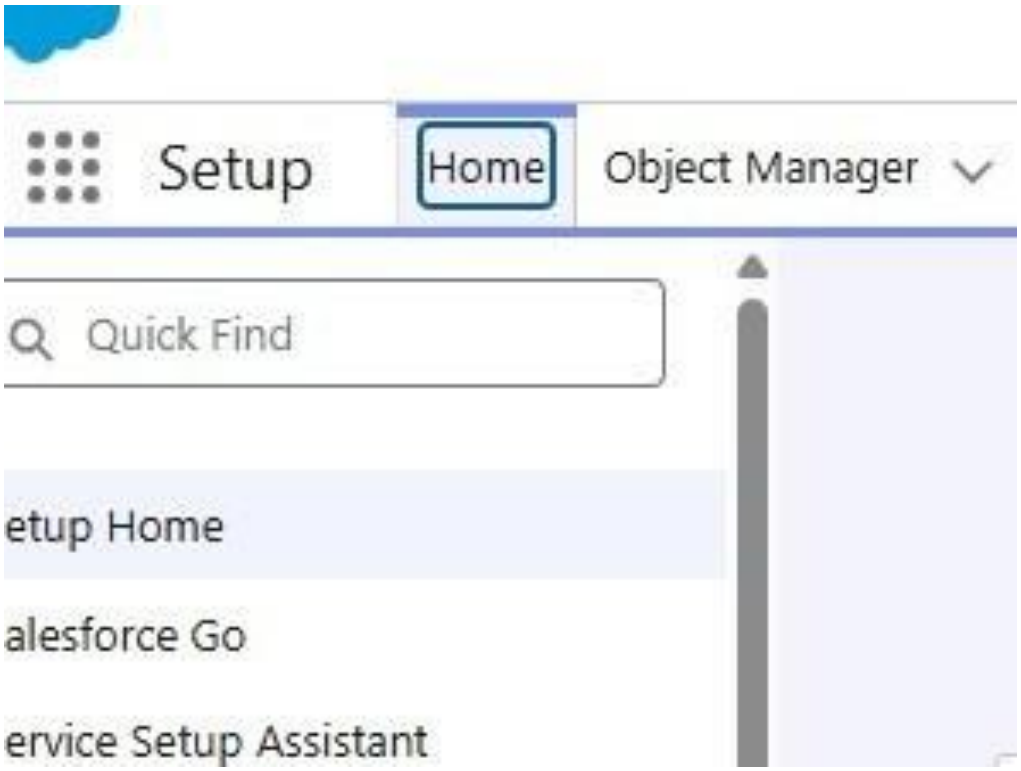
1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.

2. Click the Object Manager tab. If you don’t see it, enter Object Manager in the Object.



3. Click the Object Manager tab. If you don't see it, enter Object Manager in the Object.





1. From the setup page Click on Object Manager Click on Create Click

Label : Player
Plural Label : Players

Enter Record Name Label and Format

Record Name : Player Name

Data Type : Text

In Optional features : check the boxes forAllow Reports | Allow Activities

| Track Field History.
Search Status : check the box forAllow Search. Save

Activity 2 : Match Performance Object

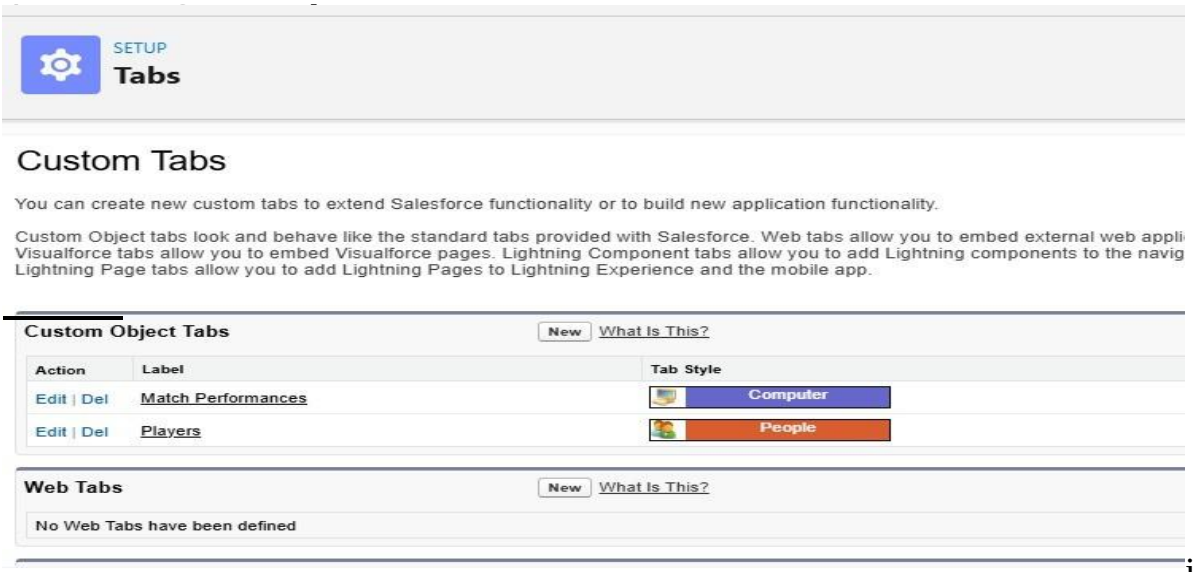
1. From the setup page Click on Object Manager Click on Create Click on

Custom Object.
Label : Match Performance
Plural Label : Match Performances Enter
Record Name Label and Format
Record Name : Customer Account Name
Data Type : Text

In Optional features : check the boxes forAllow Reports | Allow Activities

| Track Field History.
Search Status : check the box forAllow Search. Save.

Milestone 3 - Tabs



in

Milestone 4 - The Lightn

2. Fill the app name in app details and branding as follow App

   Name : CRICKSTATS

   Developer Name : Auto Populates

   Add image optional (if you want to give any image you can otherwise not mandatory) -

   Add Primary color Hex or leave it to default.
   Then click Next.

3. In App options

   Navigation style : Standard navigation

   Setup experience : setup

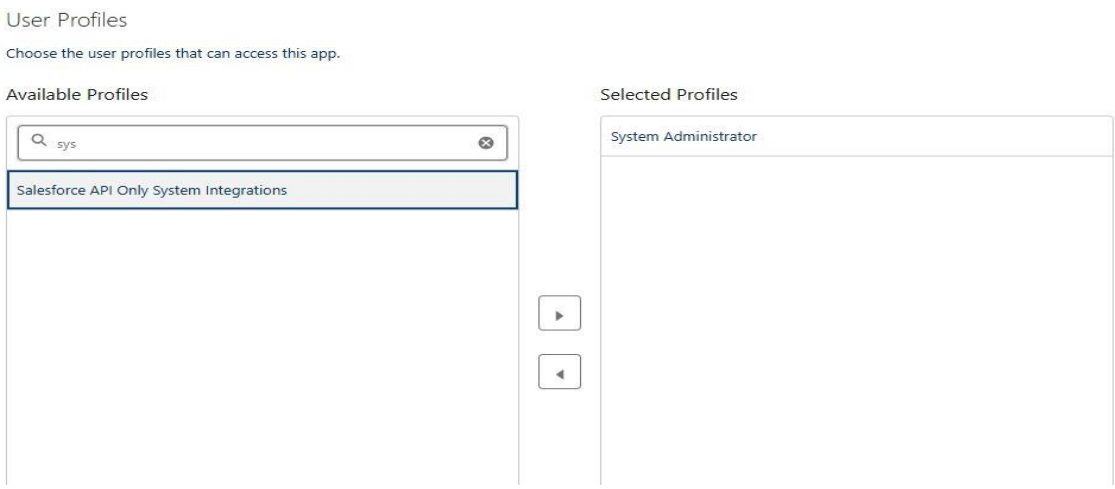
   Supporters form factors : Desktop and phone
   Then click Next

4. In Utility items
   Utility Bar alignment : Default
   Then click Next.

5. Navigation items

   Select the created Custom Objects and require standard objects Players Match Performances Reports Dashboard.

6. To Add User Profiles : System Administrator.
7. Click next.



Milestone 5 : Fields & Relationships

Activity 1 : Create Custom fields for Player object

1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.

2. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.
2. From the object manager page, In the Quick Find box, Search for the custom object you just created : Player

3. From the sidebar, click Fields & Relationships. Notice that there are already some fields there. Those are the standard fields.
4. Click New to create a custom field. Tip: Before creating a new field, do a quick

search to make sure a similar one doesn't already exist.

5. Next, choose a data type as Text. Choosing a data type helps you format the field input.

Field Name : Auto Populates

Length : 50

Click on Next Next Save and new.

9. Repeat the Same steps for remaining fields Data

Type : Text

Field

Label: Player Number

Field Name : Auto Populates

Length : 3

Click on Next Next Save and new.

Data Type : Rich Text Area

Field

Label : Profile Picture

Field Name : Auto Populates

Click on Next Next Save and new.

Data Type : Date

Field Label : Date of Birth

Field Name : Auto Populates

Click on Next Next Save and new.

Data Type : Picklist

Field Label : Role

Field Name : Auto Populates

Value : Enter value, with each value separated by a new line

Data Type : Picklist

Field Label : Batting Style

Field Name : Auto Populates

Right-Arm Medium
Left-Arm Medium

Right-Arm Spin

Left-Arm Spin

Click on Next Next Save and new.

Data Type : Picklist

Field Label : Nationality

Field Name : Auto Populates

Value : Enter value, with each value separated by a new line.

Australia

Bangladesh

England

India
New Zealand
Pakistan
South Africa
Sri Lanka
West Indies
Afghanistan
Zimbabwe
Ireland
Scotland
Netherlands
United Arab Emirates
Nepal
United States of America

Click on Next Next Save and new.

Data Type : Picklist
Field Label : Team
Field Name : Auto Populates

Value : Enter value, with each value separated by a new line

Chennai Super Kings (CSK) Delhi

Capitals (DC)

Kolkata Knight Riders (KKR)

Mumbai Indians (MI)

Punjab Kings (PBKS) Rajasthan

Royals (RR)

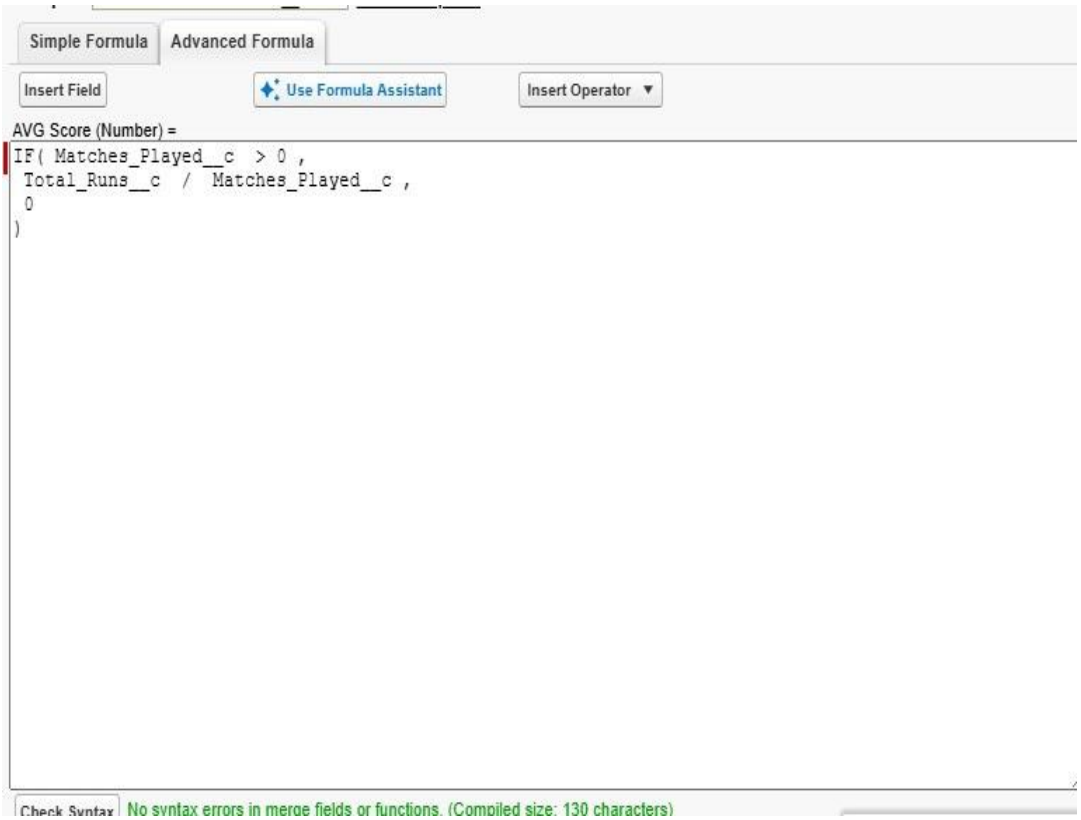
Royal Challengers Bangalore

(RCB)

Sunrisers Hyderabad (SRH)

Gujarat Titans (GT) Lucknow

SuperGiants (LSG)



Data Type : Formula Field Label : AVG Score

Field Name : Auto Populates

Formula Return : Number

Formula : IF(

Matches_Played_c > 0,

Total_Runs_c / Matches_Played_c ,

0

)

Save & New.

Activity 2 : Create Custom fields for Match Performance object.

1. Data Type : Text

   Field Label: Match Performance Name Field

   Name : Auto Populates

   Length : 80

   Click on Next Next Save and new.

2. Data Type : Checkbox

   Field Label:

Field Name : Auto Populates
Click on Next Next Save and new.

3. Data Type : Picklist
   Field Label: Match Result
   Field Name : Auto Populates

   Value : Enter value, with each value separated by a new line WON
   LOST
   DRAW
   Click on Next Next Save and new.

4. Data Type : Number
   Field Label: Runs Scored
   Field Name : Auto Populates
   Click on Next Next Save and new.

5. Data Type : Number Field
   Label: Balls Faced
   Field Name : Auto Populates
   Click on Next Next Save and new.

6. Data Type : Number
   Field Label: Fours
   Field Name : Auto Populates
   Click on Next Next Save and new.

7. Data Type : Number
   Field Label: Sixes
   Field Name : Auto Populates
   Click on Next Next Save and new.

8. Data Type : Number
   Field Label: Overs Bowled
   Field Name : Auto Populates
   Click on Next Next Save and new.

9. Data Type : Number
   Field Label: Wickets Taken
   Field Name : Auto Populates
   Click on Next Next Save and new.

10. Data Type : Number
Field Label: Runs Conceded
Field Name : Auto Populates
Click on Next Next Save and new.

11. Data Type : Number
Field Label: Maiden Overs
Field Name : Auto Populates
Click on Next Next Save and new.

12. Data Type : Number

Field
Label: Stumpings
Field Name : Auto Populates
Click on Next Next Save and new

13. Data Type : Number
Field Label: Catches
Field Name : Auto Populates
Click on Next Next Save and new.

14. Data Type : Master - Detail Relationship

Related To : Player

Field Label : Player
Field Name : Auto Populates
Click on Next Next Save.

Milestone 6 : Roll-up Summary

Activity 1 : Create Roll-up Summary for Player object.

Note :- Roll-up Summary can only be created on parent object in Master- Detail

Relationship.

1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.
2. Click the Object Manager tab. If you don't see it, enter Object
Manager in the Quick Find box.

2. From the object manager page, In the Quick Find box, Search for the custom object

   you just created : Player

3. From the sidebar, click Fields & Relationships. Notice that there are already some fields there. Those are the standard fields.
4. Click New to create a custom field. Tip: Before creating a new field, do a quick

   search to make sure a similar one doesn't already exist. Click

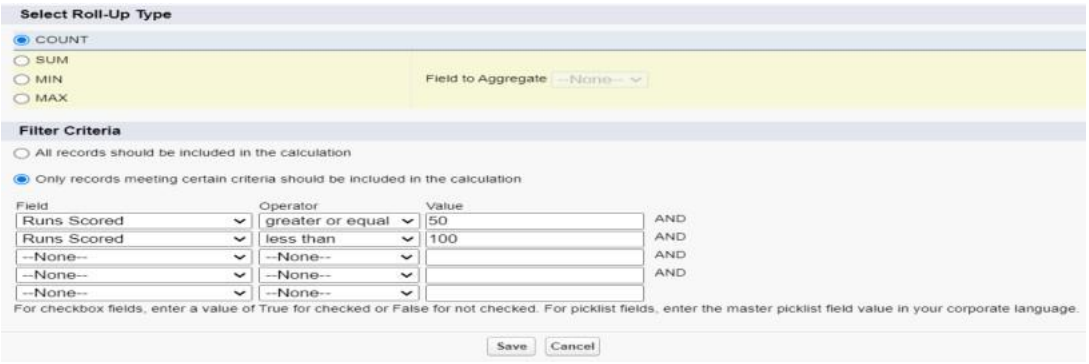
Next,

5. Choose a data type as Roll-up Summary. Click Next. Field Label: Matches Played

   Field Name : Auto Populates

   Summarized Object : Match Performances Select

   Roll-up Type : COUNT



Filter Criteria : All records should be included in the calculation.

Click on Next Next Save and new.

Data Type : Roll-up Summary

Field Label: Number of 50's

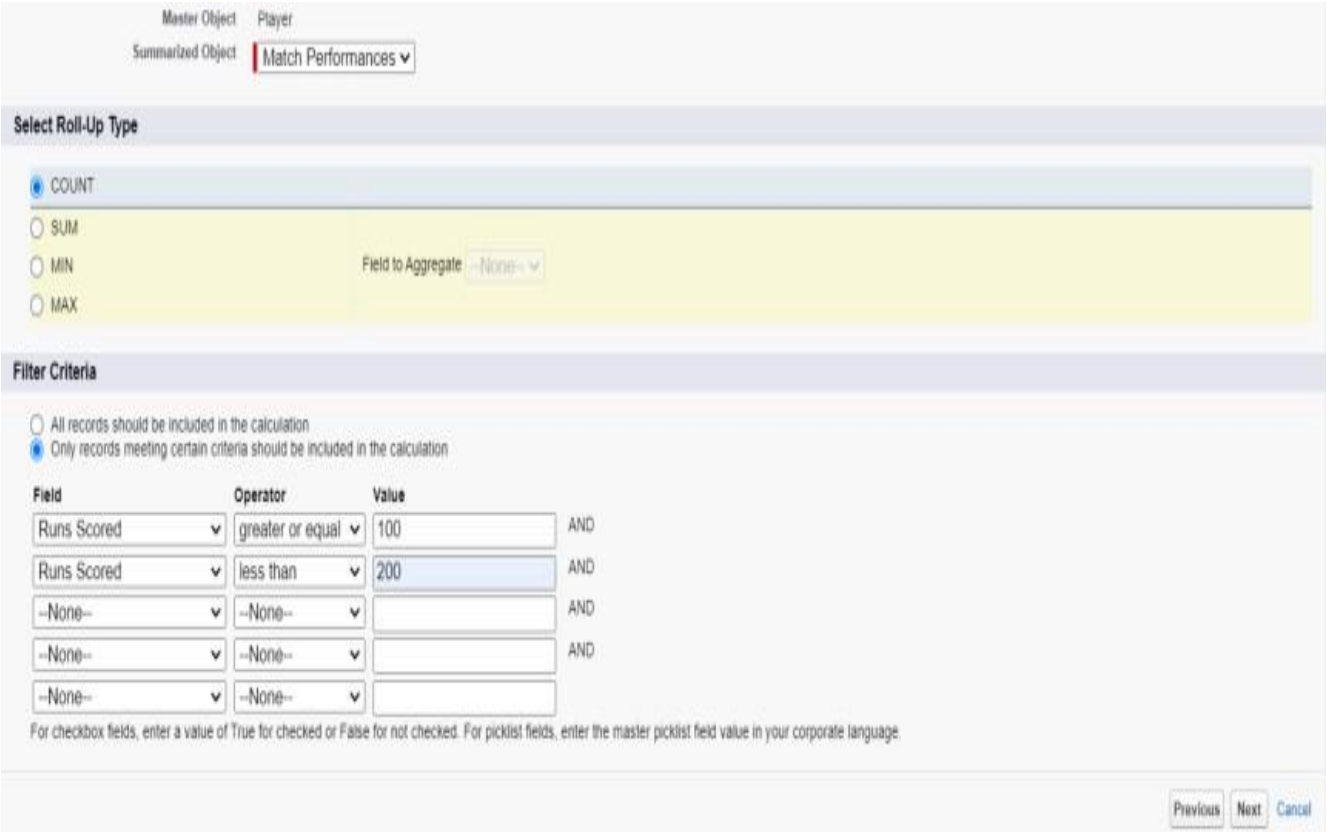
Field Name : Auto Populates

Summarized Object : Match Performances

Select Roll-up Type : COUNT

Filter Criteria : Only records meeting certain criteria should be included in the calculation.

Click on Next Next Save and new.



Summarized Object : Match Performances
Select Roll-up Type : COUNT

Filter Criteria : Only records meeting certain criteria should be included in the calculation.
Click on Next Next Save and new.

Summarized Object : Match Performances

Select Roll-up Type : SUM

Field To Aggregate : Fours
Filter Criteria : All records should be included in the calculation. Click on Next Next Save and new.

Data Type : Roll-up Summary
Field Label: Total 6's
Field Name : Auto Populates
Summarized Object : Match Performances

Select Roll-up Type : SUM

Field To Aggregate : Sixes
Filter Criteria : All records should be included in the calculation. Click on Next Next Save and new.

Data Type : Roll-up Summary

Summarized Object : Match Performances

Select Roll-up Type : SUM

Field To Aggregate : Runs Scored
Filter Criteria : All records should be included in the calculation. Click on Next Next Save and new.

Data Type : Roll-up Summary
Field Label: Total Overs Field
Name : Auto Populates

Summarized Object : Match Performances

Select Roll-up Type : SUM

Field To Aggregate : Overs Bowled
Filter Criteria : All records should be included in the calculation. Click on Next Next Save and new.

Data Type : Roll-up Summary
Field Label: Total Wickets
Field Name : Auto Populates

Summarized Object : Match Performances

Select Roll-up Type : SUM

Field To Aggregate : Wickets Taken
Filter Criteria : All records should be included in the calculation. Click on Next Next Save and new.

Data Type : Roll-up Summary Field

Label: Total Stumpings Field

Name : Auto Populates

Summarized Object : Match Performances

Select Roll-up Type : SUM

Field To Aggregate : Stumpings
Filter Criteria : All records should be included in the calculation. Click on Next Next Save and new.

Data Type : Roll-up Summary
Field Label: Total Catches
Field Name : Auto Populates

Summarized Object : Match Performances

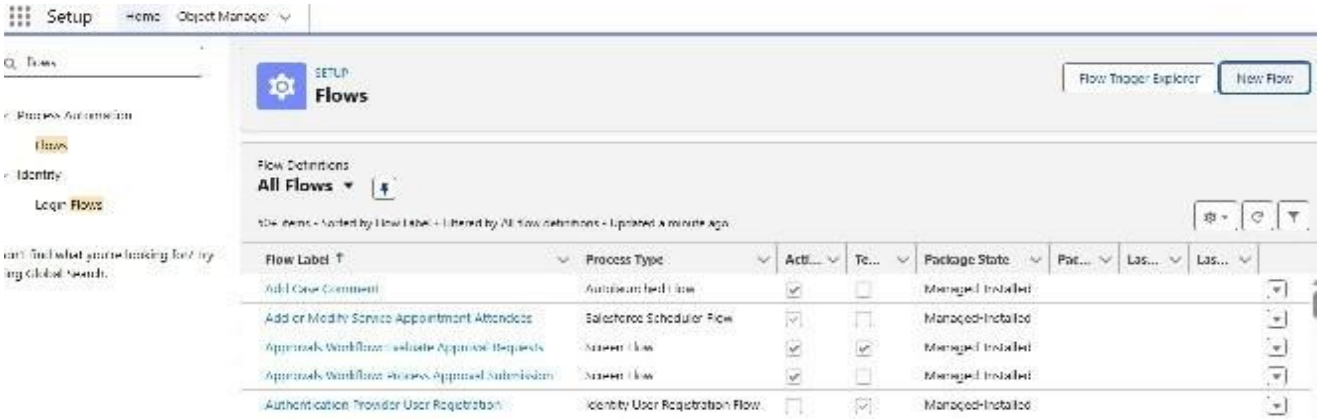
Select Roll-up Type : SUM

Field To Aggregate : Catches
Filter Criteria : All records should be included in the calculation. Click on Next Next Save and new.

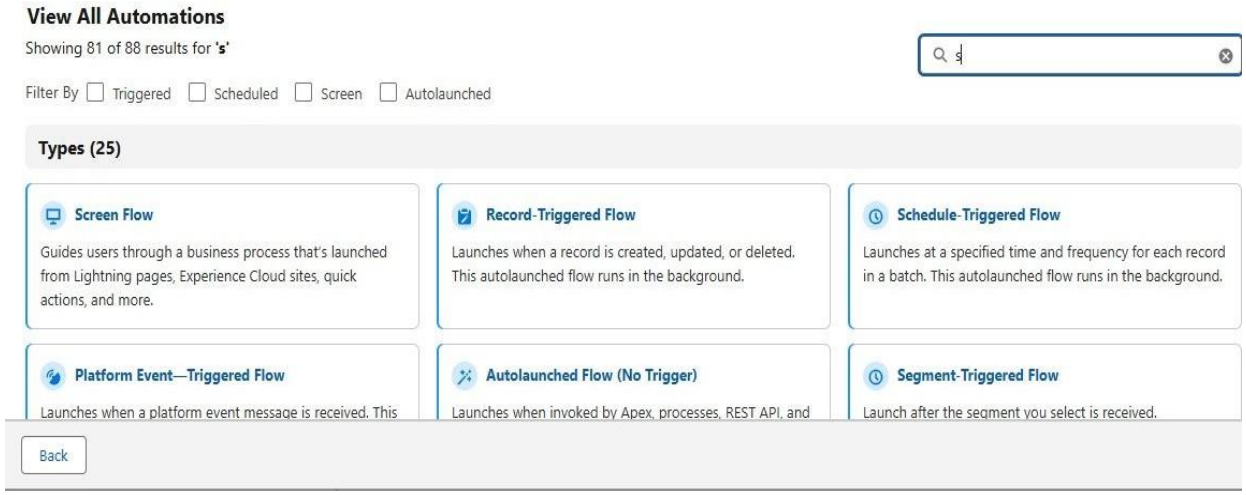
Milestone 7 : Flows

Activity 1 : Screen Flow

1. Go to setup, Type Flow in quick find box Click on the Flow and Select the New Flow.
2.



2. Select Screen Flow and click Crea



3. Click (+) Icon on th

builder and Select Screen Component.

Label : Players screen\

API : Players_screen

Select the Picklist component and place it on the screen. Label : Team

API : Team

From Component Pane on the left side select Component Type : Picklist.

Milestone 7 : Flows

Activity 1 : Screen Flow

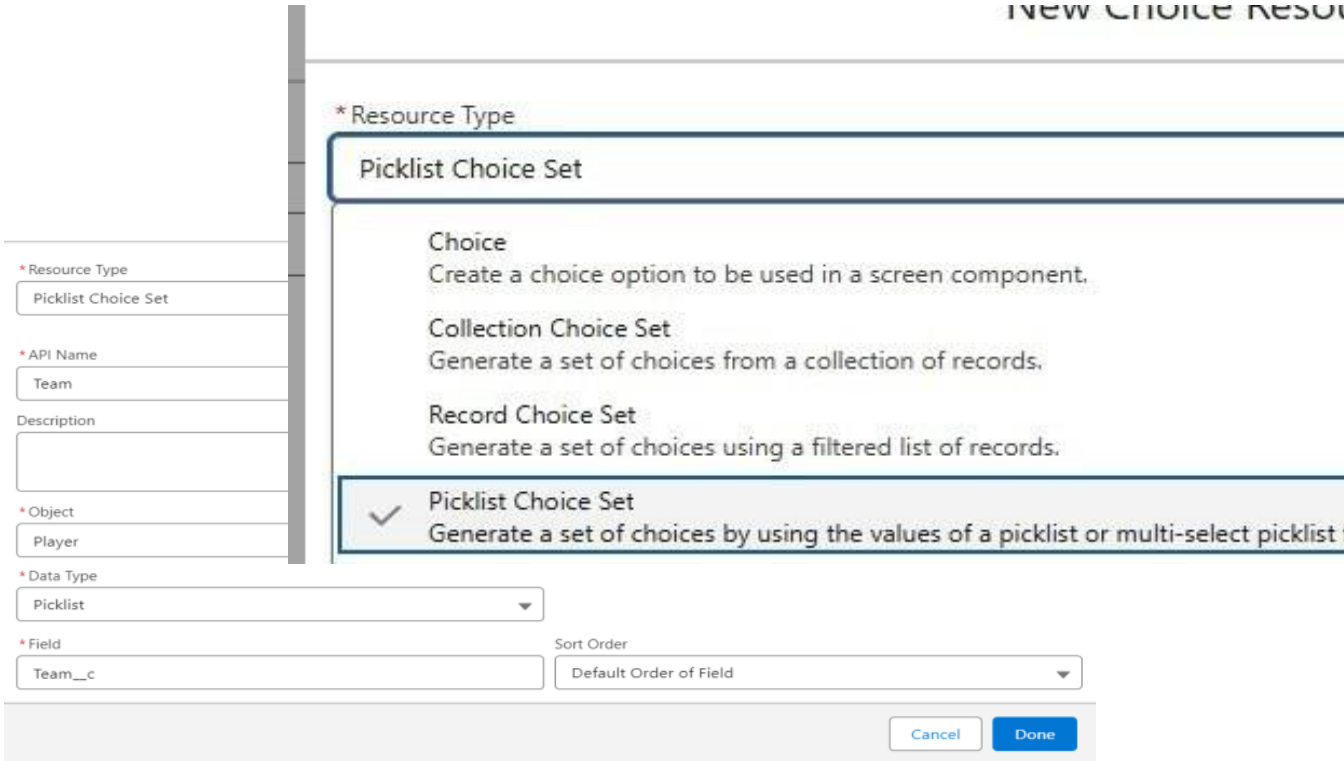
3. Go to setup, Type Flow in quick find box Click on the Flow and Select the New Flow.

Data Type : Text

Choice : Click New Choice Resources

Resource Type : Picklist Choice Set

API Name : Team



5. Click (+) Icon on the flow builder and Select Get Records Component.

Label : Players

API Name : Players
Object : Player

Value : Click New Resource
Resource Type : Variable

API Name : Dynamic

Data Type : Text

Default Value : Screen component : Team (Picklist)

ata Table

Configure Data Source : Source Collection : Players from players (Player__c)

A
P
I

N
a
m
e

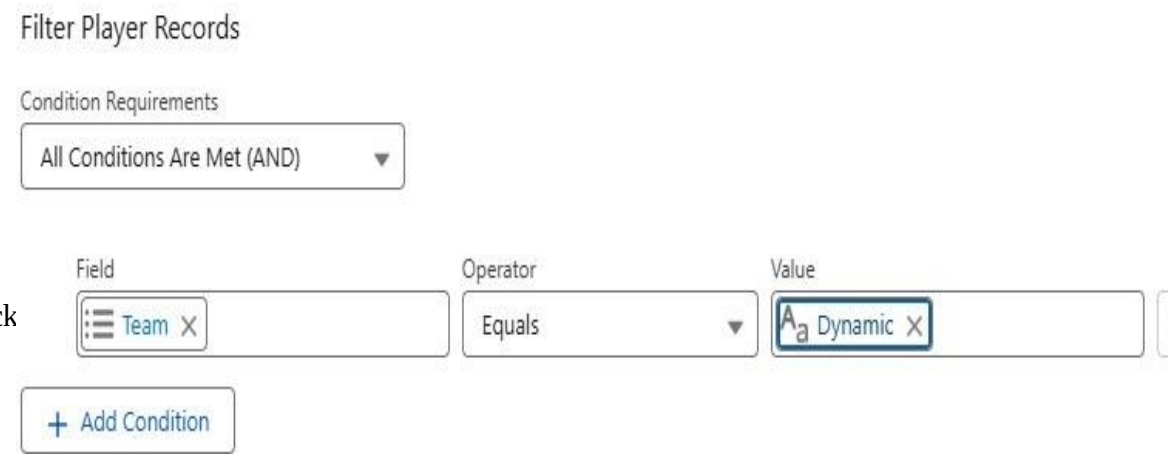
:

P
l
a
y
e
r
_
i
n
f
o
_
t
a
b
l
e

L
a
b
e
l

:

D



Click Done. Click

Save the flow

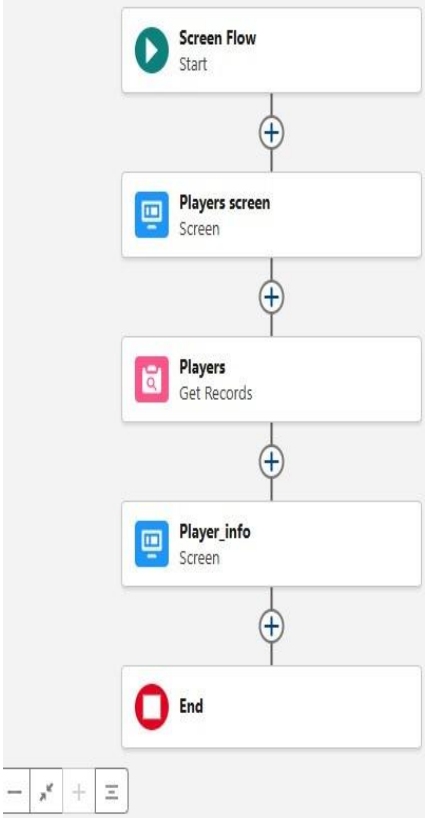
* Flow Label: Player Flow
* Flow API Name: Player_Flow

Description

Show Advanced

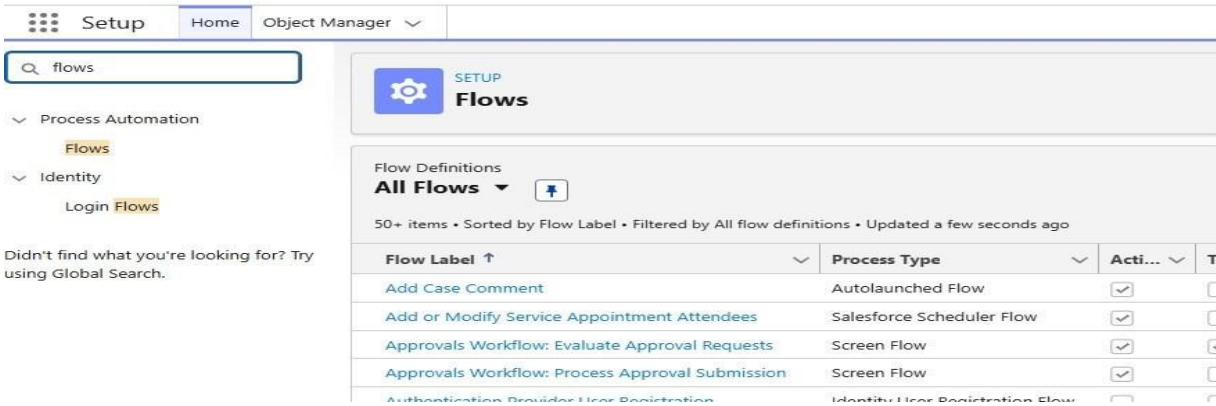
Cancel | Save

Auto-Layout | Last saved on 3/8/2026, 09:04 AM | Inactive | Run | Debug | Save As New Version | Save | Activate

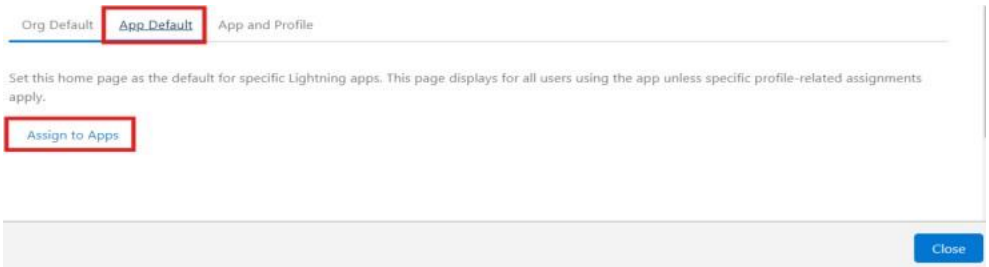


Activity 2 : Flow on Home Page

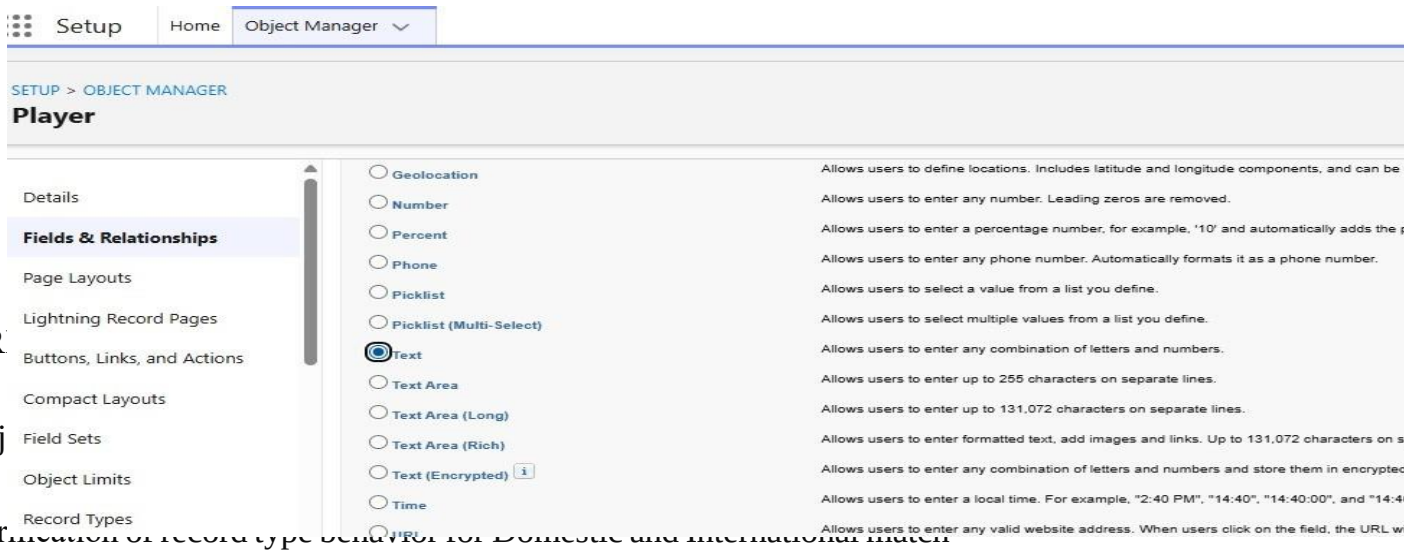
1. On your CRICKSTATS Application on the top right of the Navigation Bar click on Personalize your nav bar icon (Pencil Icon) and Add Home Page and click Save.



4. Click Save and Activation.



6. Select CRICKSTAS App and Click Next and then click Save.



CR

obj

ver

categories; and

flow execution for automatic updating of player statistics and data consistency.
The

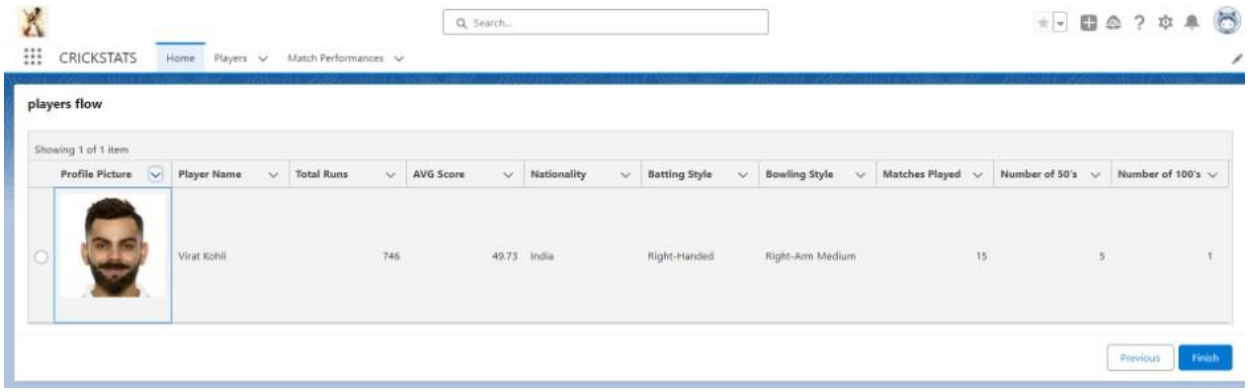
functionality of roll-up summary fields was also tested to ensure accurate calculation of

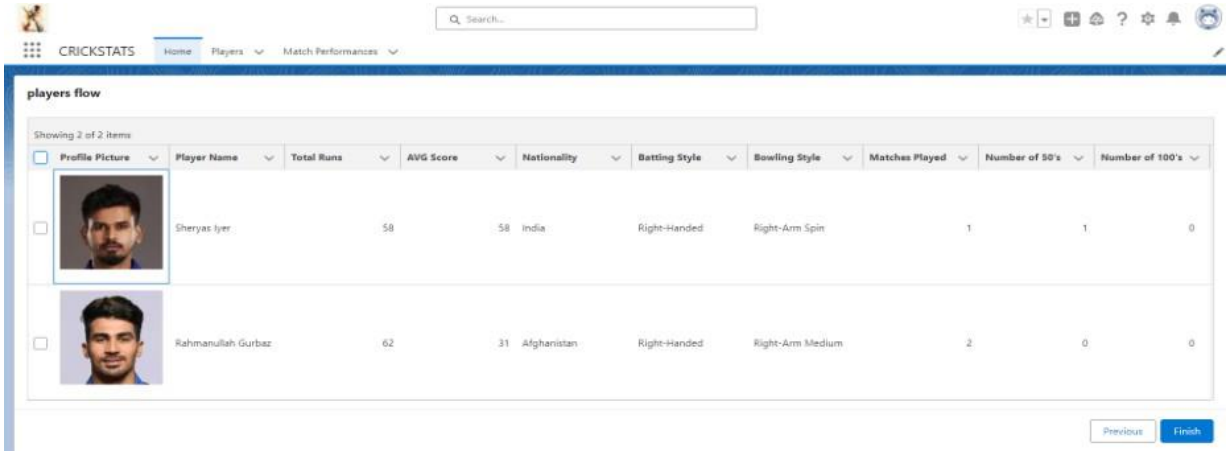
total runs, wickets, averages, and performance metrics.

In addition, system performance was evaluated by monitoring response time during data entry, report generation, and dashboard loading. The application was tested under multiple user interactions to ensure stable performance without delays.
Screenshots of

generated reports and interactive dashboards were documented to confirm accurate data visualization, real-time updates, and reliable performance tracking. These testing.
Activities ensured that the CRICKSTATS platform met both functional and performance requirements and was ready for deployment.

NOTE : Enter some records in the Player Object and Match Performance Object to get the results.





8. ADVANTAGES & DISADVANTAGES

Advantages

Centralized system

Real-time analytics

Secure platform

Scalable design

Reduced manual work

Disadvantages

Requires Salesforce license

Internet dependency

Limited offline access

9. Conclusion:

CRICKSTATS successfully implements a cloud-based cricket performance tracking system using Salesforce. The system automates data management, improves reporting accuracy, and supports informed decision-making.

The project demonstrates practical application of SDLC, Agile methodology, and CRM-based development.

10. FUTURE SCOPE

Mobile application integration

AI-based performance prediction.

Player ranking system

Public statistics portal

Multi-league support

11. APPENDIX

Salesforce Objects & Fields
Manual match statistics Flow
Configurations

Thank You